Probabilistic Machine Learning

Lecture 03: Generative Models and Naive Bayes

Prof. Dr. Michael Granitzer

Where We Left Off

Lecture 02: 100 students, a count table.

	PML	DB	total
fail	10	6	16
pass	30	54	84
total	40	60	100

$$P(\text{PML} \mid \text{fail}) = \frac{P(\text{fail} \mid \text{PML})\,P(\text{PML})}{P(\text{fail})} = \frac{10}{16}$$

One hypothesis, **one** piece of evidence: Bayes' theorem reads the table backwards.

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Now the hypothesis is a class — fraud or honest — and the evidence a whole transaction. What changes?

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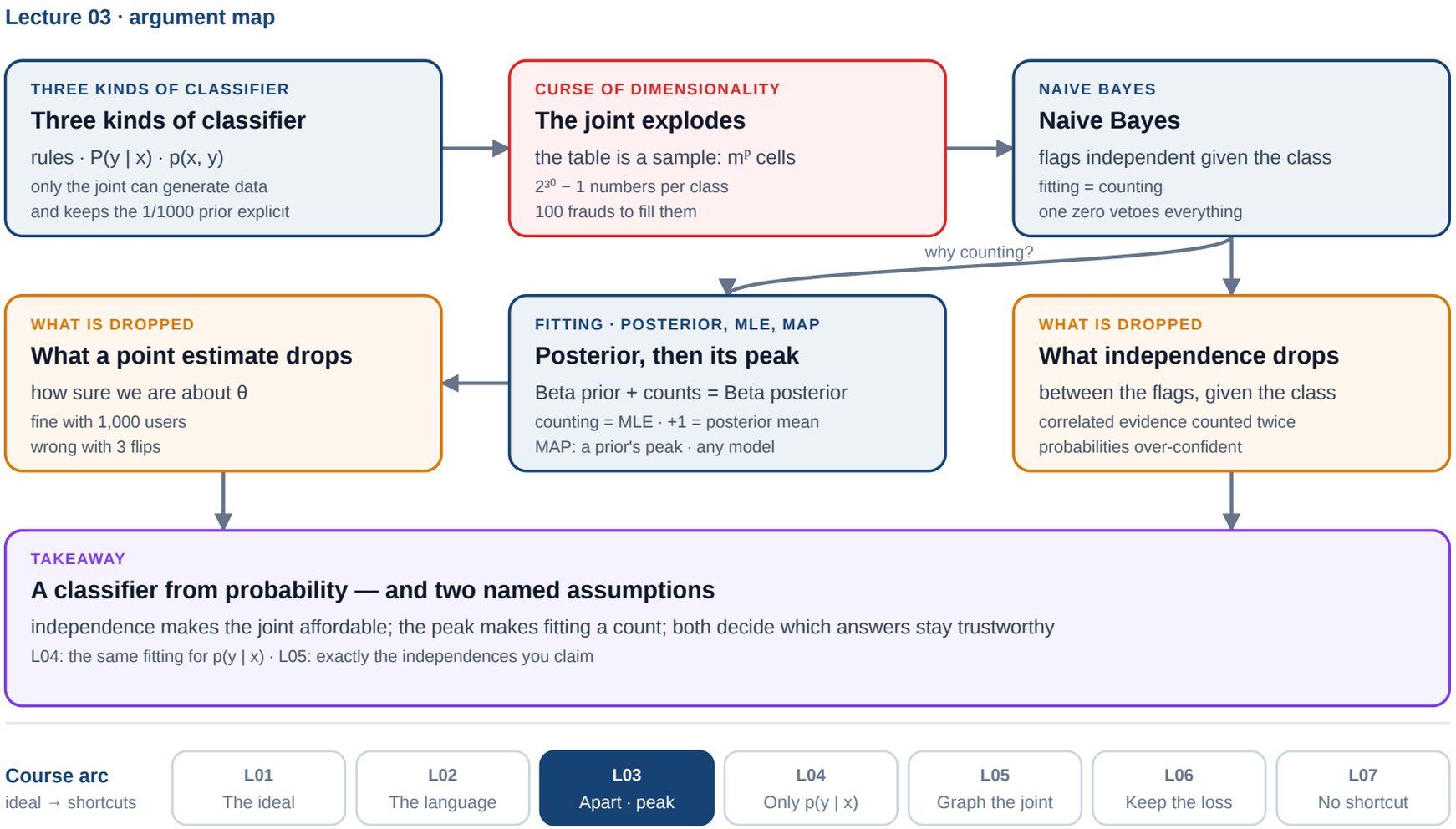
Probabilities at all? A rule *if amount > 1k € and new device, then fraud* also classifies.

Today: Naive Bayes and Its Two Assumptions

A classifier built from Bayes' theorem needs two assumptions to become estimable — rows 1 and 2 of Lecture 01's table of shortcuts:

assumption	replaces	what is lost	module
Conditional independence	the joint $p(x \mid y)$ by $\prod_j p(x_j \mid y)$	the dependences between attributes	Naive Bayes
Point estimate (MLE, MAP)	the posterior over a parameter by its peak	how sure we are about the parameter	Fitting the Parameters

Argument Map



Three Kinds of Classifier

Guiding Question: A rule, a model of $P(y \mid x)$, a model of how the data arise — what can each one tell us about a transaction, and what does the most informative one cost?

Data and Hypothesis as Random Variables

amount	distance	…	fraud?
49.90 €	3 km	…	no
1,250 €	900 km	…	**yes**
X_1	X_2	…	Y

- **Data** $x = (x_1, \ldots, x_p)$: the values of the attributes $X_1, \ldots, X_p$ — what we observe.
- **Hypothesis** y: the value of the class variable Y — what we want to know.
- **Decision:** the posterior $P(y \mid x)$.

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The student table had hypothesis A = "PML" and evidence B = "fail". Which of the two is the data, which the hypothesis?

Parameters θ

A **model** is a family of distributions $p(\,\cdot\,;\theta)$. The **parameters** θ pick one member; training sets them.

Read the notation

$p(x \mid y;\theta)$ — the probability of data x, **given** the hypothesis y, **for** the parameter value θ.

The bar conditions on a random variable, the semicolon fixes a parameter.

Example

amount given fraud ~ normal: $\theta = (\mu, \sigma)$.

Fraud share $P(Y = \text{fraud};\theta) = \theta$: one number.

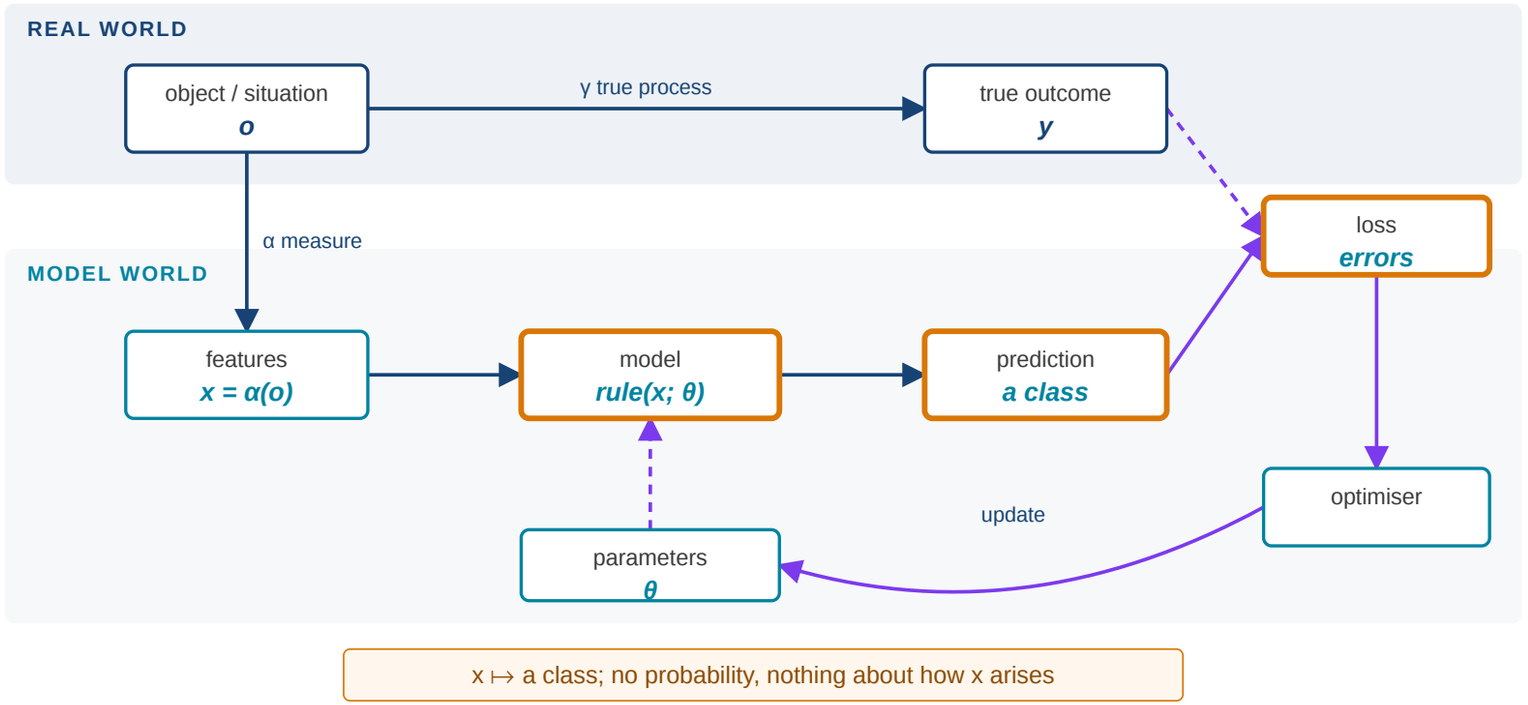
Rule-Based Classifier

A **rule-based classifier**: rules "if $C_k(x)$ then c_k" plus a default class. The **rules are the parameters** θ.

$x_1 > a,\ x_2 > b \rightarrow$ fraud, else honest: $\theta = (a, b)$, trained for the fewest errors

$$\hat{\theta} = \arg\min_{\theta} \sum_{i} 1[\hat{y}_i \neq y_i]$$

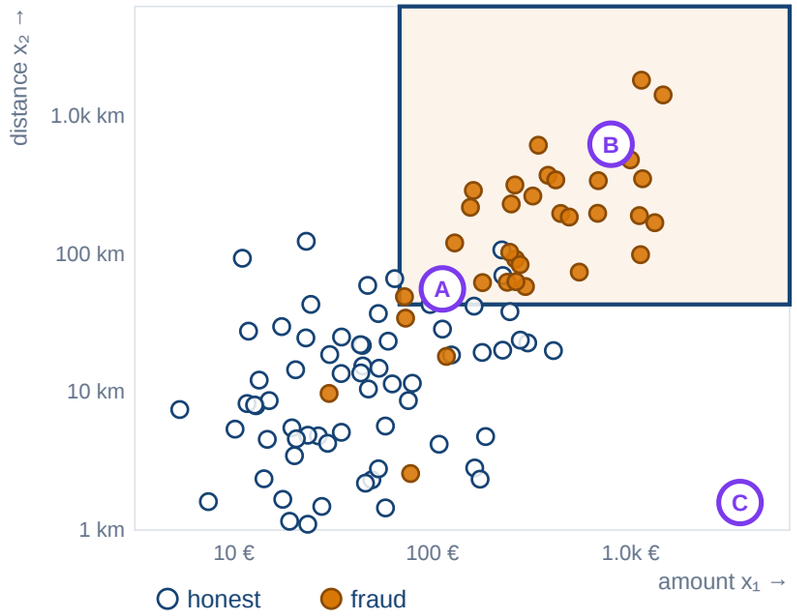
Discriminant function: rules



x ↦ a class; no probability, nothing about how x arises

Rules From Data

A and B get the same answer; C gets one without any warning



Learned rule

if amount > 68 €
and distance > 43 km
then fraud, else honest

training errors: 6 of 99

fraud share in the box: 94%
— the only "score" the rule has

A: fraud, 94%
just inside the box

B: fraud, 94%
deep inside the box

C: honest
5,000 € at 1 km — no such row

Which transaction is the rule surest about — A, B or C?

What a Rule Can Not Say

- ✗ **How sure.** One share for the whole box — A and B alike.
- ✗ **Honestly.** The share is measured on the rows that chose the box: optimistic, 100% for a box searched until pure.
- ✗ **Under a new base rate.** The training mix is baked into the thresholds.
- ✗ **That an input is strange.** No model of x: C gets an answer like any other.

Examples

- **Fraud rule engines:** "amount > 1k € and new device → block".
- **Decision trees:** rules learned by splitting, one class per leaf.

A rule is a **discriminant function**: $x \mapsto$ a class. No distribution is modelled.

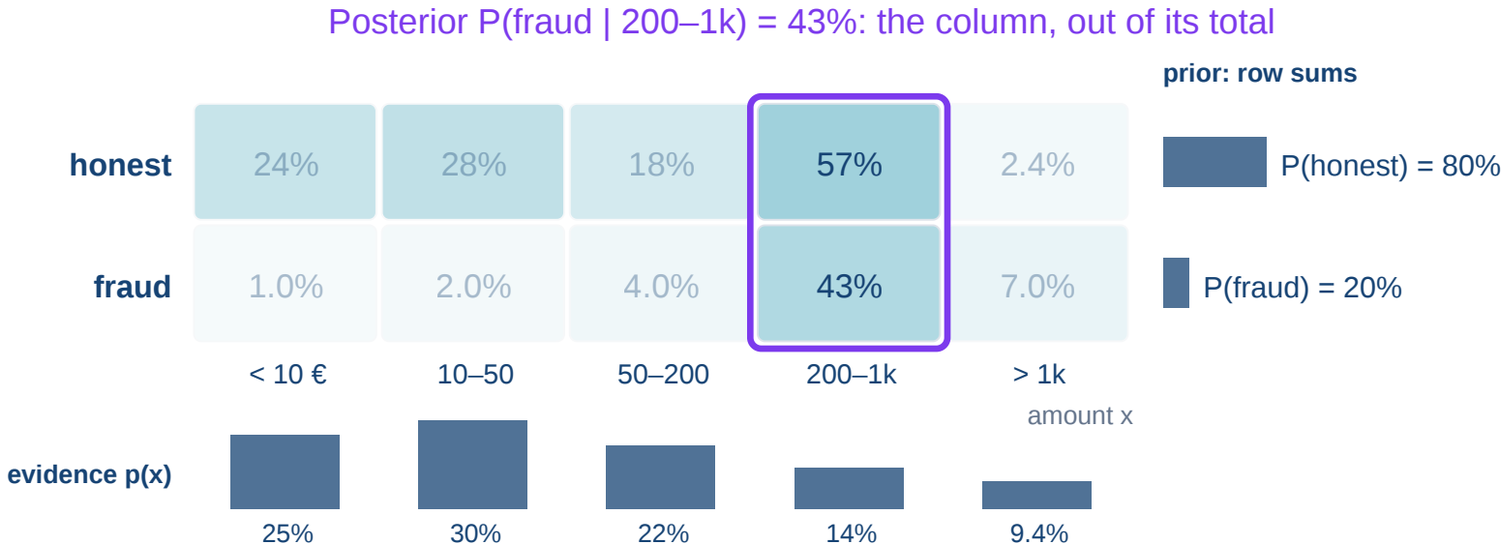
The Joint Distribution $p(x_1, \ldots, x_p, y)$

A **probabilistic** classifier starts from the joint distribution of data and hypothesis. Every question is read off it:

question	what is read off the joint
Is this transaction fraud?	$P(y \mid x) = p(x, y) \,/\, p(x)$
How does fraud look?	$p(x \mid y) = p(x, y) \,/\, P(y)$
How common is fraud?	$P(y) = \sum_x p(x, y)$
How unusual is this transaction?	$p(x) = \sum_y p(x, y)$
The distance is missing — now?	sum x_2 out, then condition

> The joint is the course's ideal: model everything, then condition.

Discriminative: Model $P(y \mid x; \theta)$ Only



Fit **exactly** the decision: a distribution over y for every x.

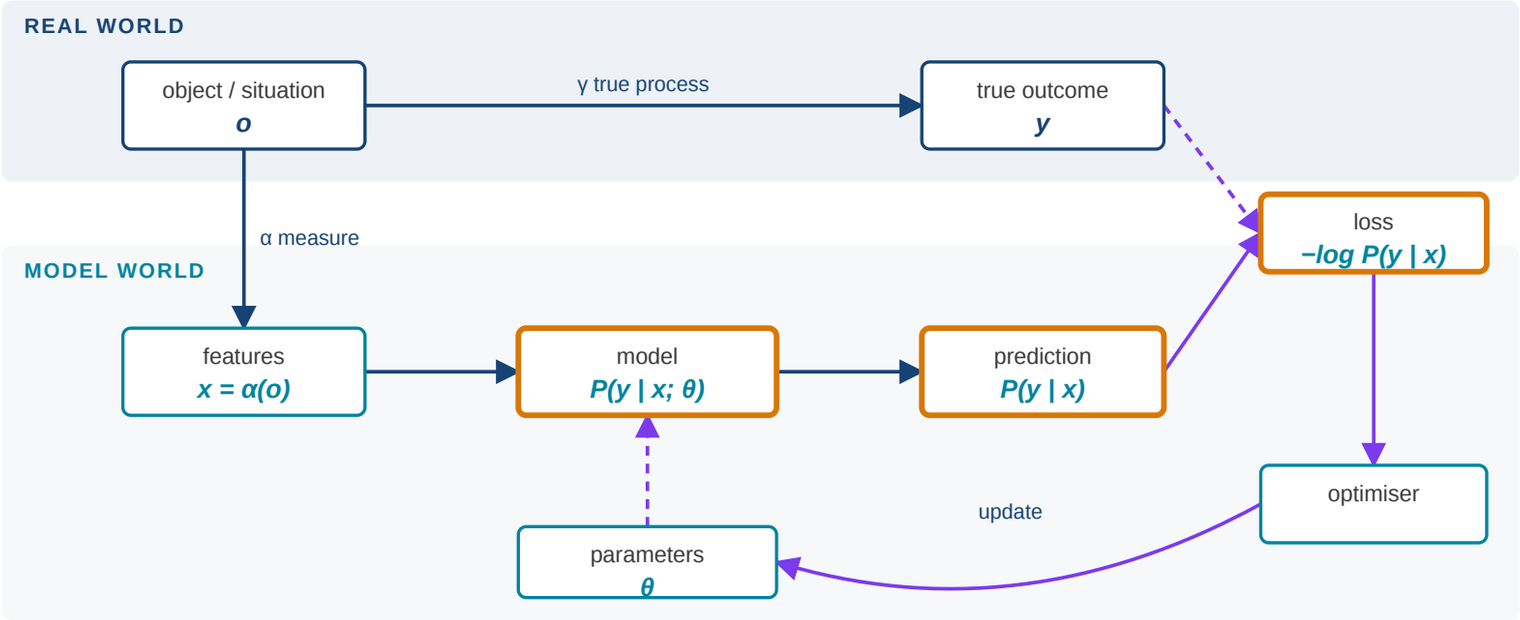
- ✗ no $p(x)$, no $p(x \mid y)$
- ✗ no Bayes: nothing to turn around

Examples

Logistic regression · neural networks with a softmax output

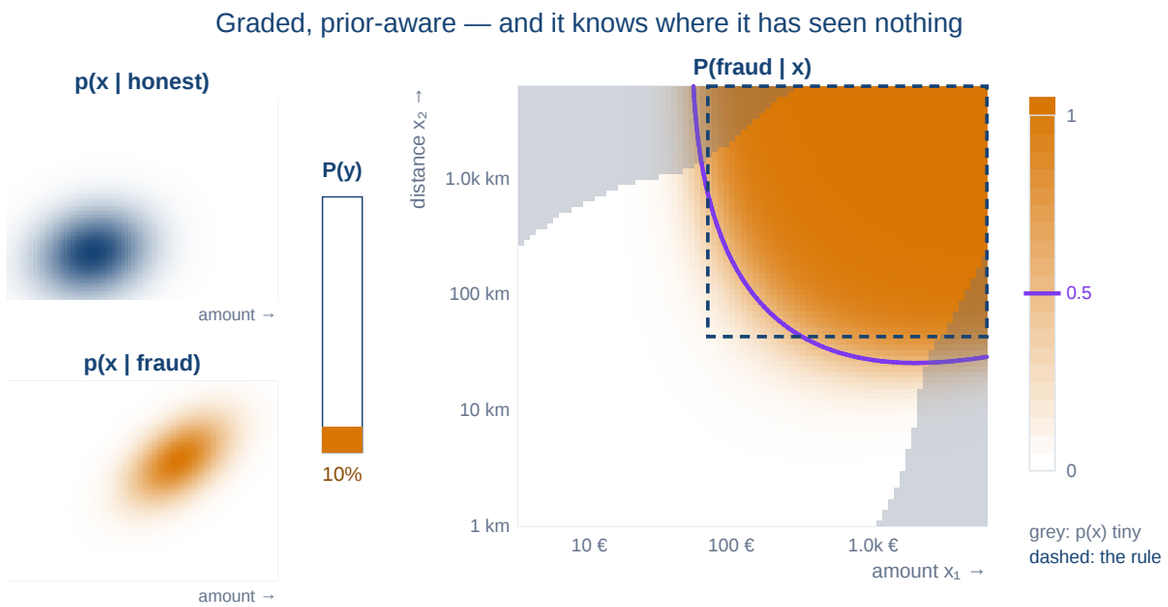
The Discriminative System in the Learning System

Discriminative model: p(y | x) only



a distribution over y for each x; x itself is never modelled

Generative: Model $p(x \mid y;\theta)\,P(y;\theta)$



Set $P(\text{fraud})$ to 0.1%. What happens to the boundary — and why can the rule not do this?

Generating Data

Run the model forwards: draw $y \sim P(y)$, then $x \sim p(x \mid y)$ — a new data point.

Naive Bayes spam filter

Words drawn one by one from $p(\text{word} \mid \text{spam})$: *free winner click offer meeting now* — each word fits, the sentence does not.

Language models

$p(x_1, \ldots, x_T) = \prod_t p(x_t \mid x_{<t})$, fitted by maximum likelihood, sampled word by word.

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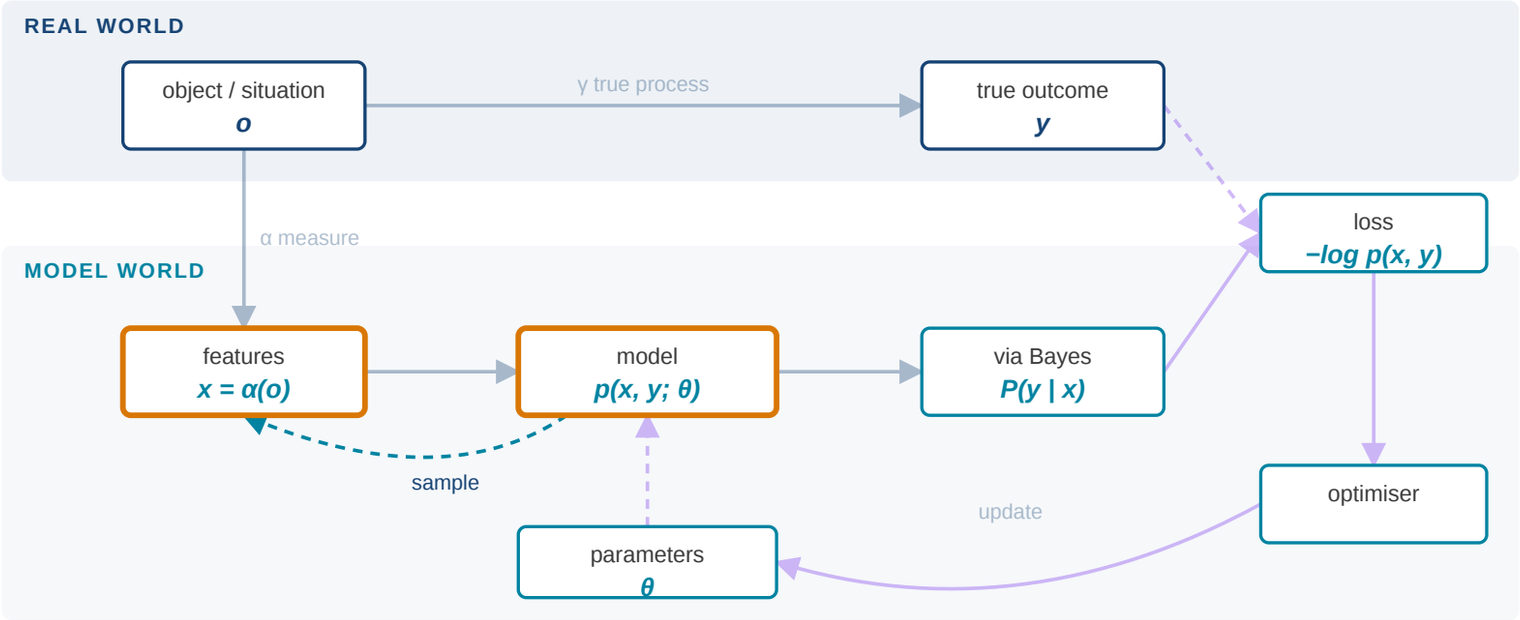
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Neither rules nor $P(y \mid x)$ can do this: generating needs a model of x.

The Generative System in the Learning System

Generative model, sampling



draw y, then x from p(x | y): a new data point the model would produce

Three Kinds of Classifier

	rules	discriminative	generative
models	$x \mapsto$ class	$P(y \mid x; \theta)$	$p(x \mid y; \theta)\, P(y; \theta)$
how sure	✗	✓	✓
new base rate	✗ refit	correct from outside	✓ swap $P(y)$
unusual input	✗	✗	✓ $p(x)$ small
missing field	✗ ad hoc	✗ ad hoc	✓ sum it out
generate data	✗	✗	✓
examples	decision tree	logistic regression	naive Bayes, language model

The price of the right column: a distribution over **all** attributes, per class.

Bayes' Theorem With Two Red Flags

Two flags at once, D = new device and F = foreign IP: the evidence is a **conjunction** of events.

$$P(\text{fraud} \mid D \cap F) = \frac{P(D \cap F \mid \text{fraud}) \cdot P(\text{fraud})}{\sum_y P(D \cap F \mid y)\, P(y)}$$

Among frauds, 60% show D and 70% show F. How many show both?

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1. **Not fixed by the parts.** Anything from 30% to 60% fits; the data say 55%. $P(D \cap F \mid y)$ is a number of its own.
2. **One number per combination.** p flags: $2^p - 1$ per class.
3. **Ever rarer.** Each added flag shrinks the count behind the conjunction.

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> Multiplying the parts, $0.60 \cdot 0.70 = 0.42$, is correct only if D and F are **independent given the class** — the assumption of **naive Bayes**.

The Training Table Is a Sample

amount (€)	country	hour	new device?	… 30 flags	fraud?
49.90	DE	14	no	…	no
1,250.00	BR	3	yes	…	**yes**
12.50	DE	19	no	…	no

Each column is a **random variable** X_j, each row one **draw** from $p(x, y)$. The fraud rows are draws from $p(x \mid \text{fraud})$.

Estimate $p(x \mid \text{fraud})$ by counting: of 100 frauds, how many look exactly like the next transaction?

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> Almost always **none**. Counting needs examples of every combination of values.

The Joint Explodes

30
yes/no red flags per transaction

→

$2^{30} - 1$
≈ 1.07 billion numbers for $p(x \mid \text{fraud})$ alone

→

100
frauds among 100,000 transactions to estimate them

Almost every cell of the table is **empty**. Its count is 0, so its maximum likelihood estimate is 0 — every fraud that looks slightly new becomes "impossible".

The Curse of Dimensionality

attributes	value combinations (10 values each)	frauds per combination (100 frauds)
1	10	10
2	100	1
3	1,000	0.1
30	10^{30}	10^{-28}

Every attribute **multiplies** the cells and **divides** the data per cell. Counting needs data in every cell — a full joint over many attributes can never be estimated by counting.

Two Ways Out

Simplify the structure

Assume attributes are **independent given the class**: one small table per attribute instead of one huge joint.

30 flags: **30** numbers per class, not $2^{30} - 1$.

NAIVE BAYES — TODAY; GRAPHS — LECTURE 05

Use a parametric family

Replace a table of counts by a **formula with few parameters**.

amount given fraud ~ log-normal(μ, σ): **2** numbers, not 100 bins.

LECTURE 02

Both are **assumptions**. They make the model estimable — and wrong wherever the world disagrees with them.

Mini-Exercise

A fraud model uses only **three** yes/no flags.

How many numbers does the generative model need — both classes and the class prior together?

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15. Each class needs a full table over $2^3 = 8$ flag patterns: 7 free numbers (they sum to 1). Two classes give 14, and $P(\text{fraud})$ adds one.

Takeaway

The generative route models **how the data arises** — $p(y)$ and $p(x \mid y)$ — and converts with Bayes. Counting that joint needs data in every combination of values, which **explodes** with the number of attributes. The way out is an assumption: **simpler structure**, **parametric families**, or both.

Naive Bayes

Guiding Question: Its central assumption is almost always false, and it classifies well anyway. How can both be true?

Conditional Independence

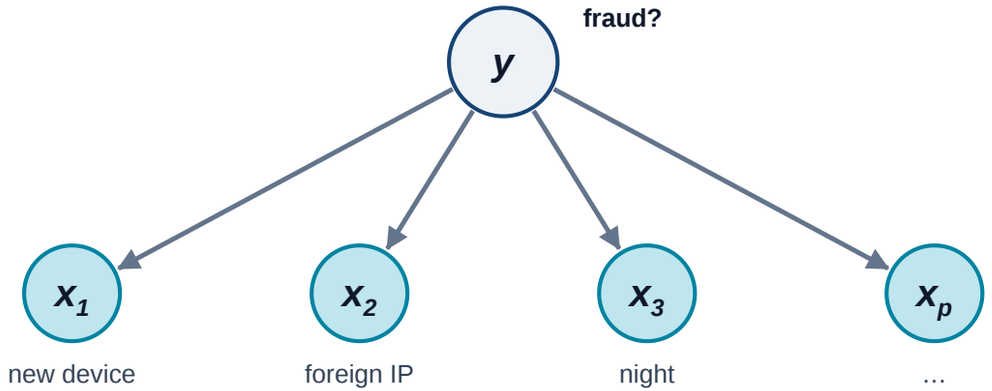
Definition. A and B are **conditionally independent given** C, written $A \perp\!\!\!\perp B \mid C$, iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C) \quad \Longleftrightarrow \quad P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once C is known, B tells us nothing more about A.**

Among frauds only: does a new device make a foreign IP more likely?

The Naive Assumption



$p(y) \cdot p(x_1 \mid y) \cdot p(x_2 \mid y) \cdot \ldots \cdot p(x_p \mid y)$

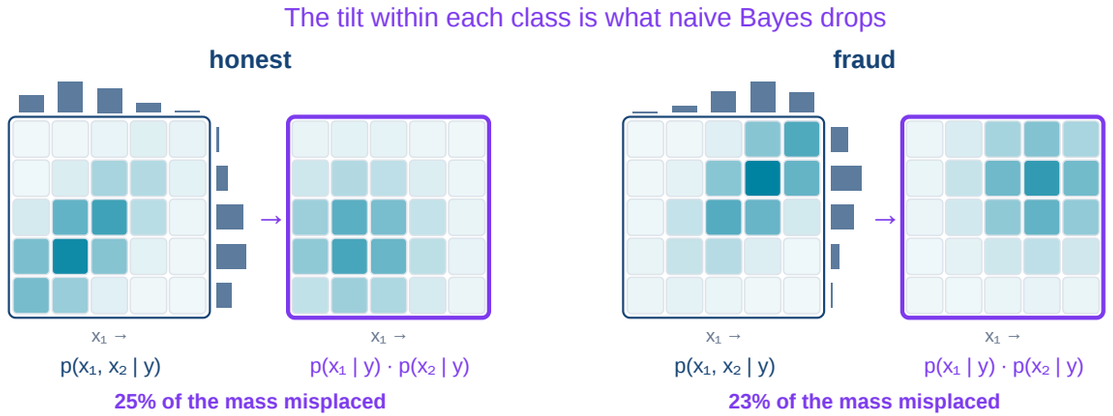
Assume the attributes are **independent given the class**:

$$p(x_1, \ldots, x_p \mid y) = \prod_{j=1}^{p} p(x_j \mid y)$$

flags	full joint, per class	naive
3	7	3
30	1,073,741,823	30

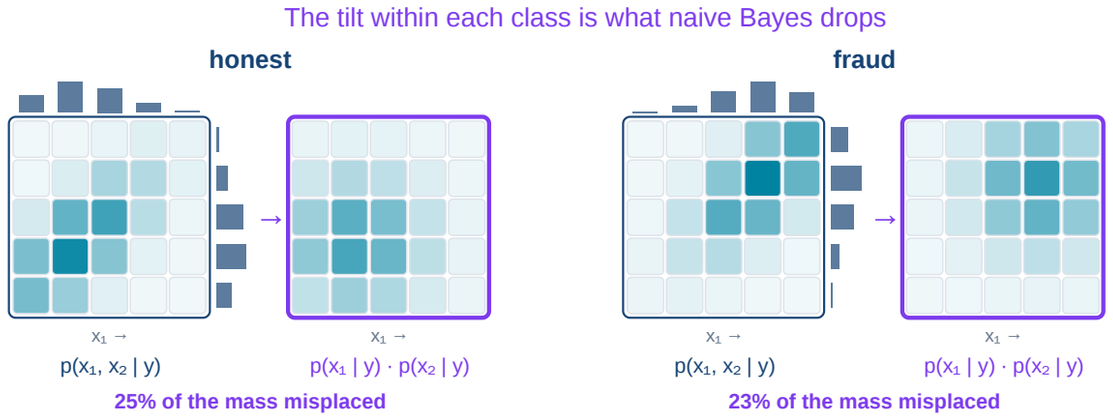
Shortcut 2 — assume it apart. Dropped: every dependence *within* a class.

Assuming It Apart, Drawn



Predict: set the correlation to 0. What happens to the two pictures?

Assuming It Apart, Drawn



Predict: set the correlation to 0. What happens to the two pictures?

They become **identical**. Naive Bayes is exact exactly when the attributes are independent **within** each class.

Fitting Is Counting

$$\hat{P}(y) = \frac{n(y)}{n} \qquad \hat{P}(x_j = v \mid y) = \frac{n(x_j = v,\ y)}{n(y)}$$

One small table per attribute

Each attribute, within one class, is a **categorical** variable — for a yes/no flag, a Bernoulli (Lecture 02).

Estimated by its share

$P(\text{new device} \mid \text{fraud})$: of the 100 frauds, how many came from a new device? 60, so 0.6.

Naive Bayes is fitted in **one pass over the table**, with no optimiser. Why a share is the right estimate — and when it is not — is the last module.

The Classifier

$$P(y \mid x_1, \ldots, x_p) = \frac{P(y) \prod_j p(x_j \mid y)}{\sum_{y'} P(y') \prod_j p(x_j \mid y')}$$

For two classes, divide the two posteriors — the normaliser cancels:

$$\underbrace{\frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)}}_{\text{posterior odds}} = \underbrace{\frac{P(\text{fraud})}{P(\text{honest})}}_{\text{prior odds}} \cdot \prod_{j=1}^{p} \underbrace{\frac{p(x_j \mid \text{fraud})}{p(x_j \mid \text{honest})}}_{\text{likelihood ratio } \mathrm{LR}_j}$$

> Every observed attribute **multiplies the odds** by its likelihood ratio: above 1 it speaks for fraud, below 1 against.

Three Red Flags

100,000 transactions, 100 of them fraud: prior odds $1 : 999$.

flag	among frauds	among honest	likelihood ratio
new device	60%	10%	6
foreign IP	70%	5%	14
night	50%	20%	2.5

A transaction shows all three flags. Is it more likely fraud than not?

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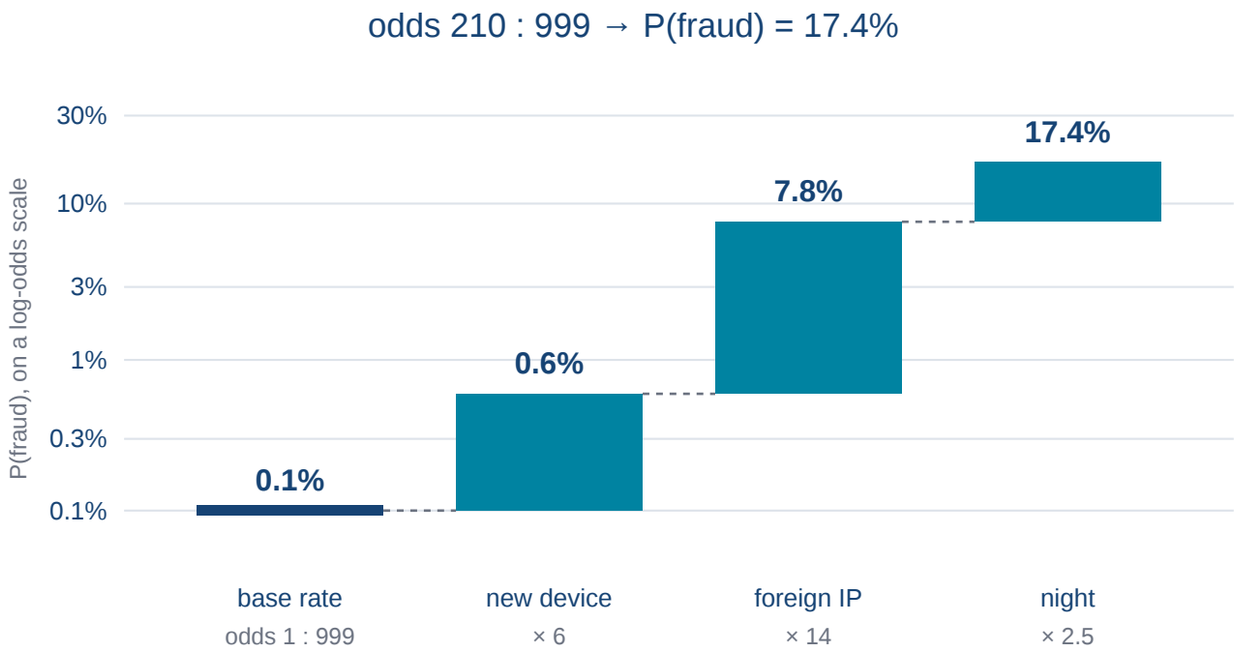
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A transaction shows all three flags. Is it more likely fraud than not?

$$\text{odds} = \frac{1}{999} \cdot 6 \cdot 14 \cdot 2.5 = \frac{210}{999} \quad \Longrightarrow \quad P(\text{fraud} \mid x) = \frac{210}{1{,}209} \approx 17\%$$

> Three red flags, each one strong — and still **mostly honest**. The base rate of Lecture 02, once more.

Evidence Adds Up



Predict before each click: which flag moves the probability most?

One Zero Vetoes Everything

A new flag: **payment to a crypto exchange**. Among the 100 frauds it never occurred.

$$\hat{P}(\text{crypto} \mid \text{fraud}) = \frac{0}{100} = 0$$

A transaction with this flag gets posterior odds **exactly 0** — whatever the other thirty flags say. One unseen value vetoes all other evidence.

COUNTING, FAILS

"Not seen in 100 frauds" became "impossible for a fraud" — a share of a small count, read as certainty.

Laplace Smoothing

Laplace smoothing pretends each of the k_j values was seen **once more**:

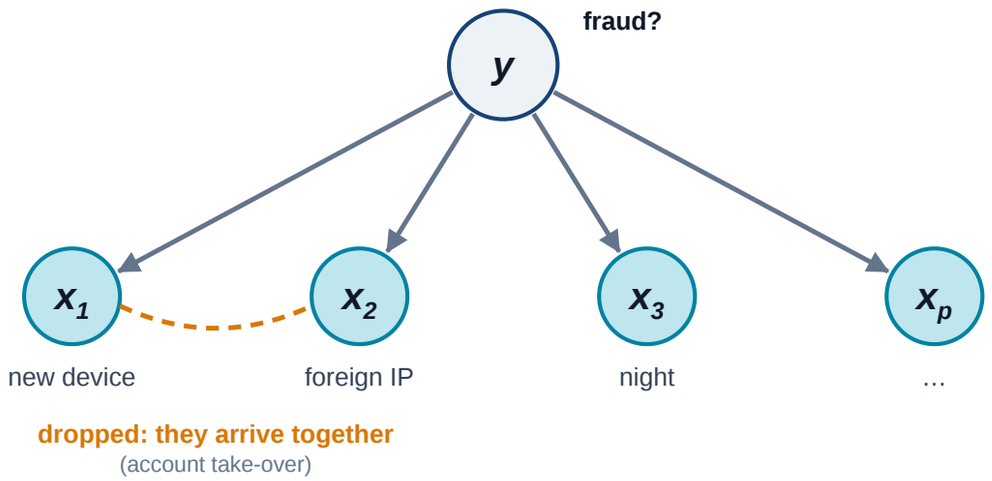
$$\hat{P}(x_j = v \mid y) = \frac{n(x_j = v,\ y) + 1}{n(y) + k_j}$$

For a yes/no flag ($k_j = 2$) this is $\frac{h+1}{n+2}$.

count among 100 frauds	share	smoothed
0	0	$1/102 \approx 0.010$
60	0.60	$61/102 \approx 0.598$

> Where the count is large, nothing changes; where it is 0, the veto is gone. Why **+1**, and not +5? The next module: smoothing is a **prior**.

Counting Evidence Twice



In an account take-over, a new device and a foreign IP **arrive together**. Suppose both appear on 55% of frauds and 4% of honest transactions:

	ratio for the pair	$P(\text{fraud})$ with night
naive Bayes	$6 \cdot 14 = 84$	17.4%
the real pair	$0.55/0.04 \approx 14$	3.3%

Correlated flags are **counted twice**, so naive Bayes is **over-confident**. Its ranking of transactions often survives; its probabilities do not.

Mini-Exercise

A bug feeds the **foreign-IP** flag into the model twice.

Does the fraud probability of our three-flag transaction go up, down, or stay — and what does that say about correlated attributes?

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Up — from 17% to about 75%. The odds get a second factor 14: $2{,}940 : 999$. Naive Bayes cannot tell a copy from new evidence; a strongly correlated attribute is almost a copy.

Takeaway

Naive Bayes **assumes the joint apart** — attributes independent given the class — and fits every factor by **counting**, smoothed by pseudo-counts so no zero can veto. It is cheap, needs little data and ranks well. Its probabilities are over-confident wherever attributes move together.

Fitting the Parameters: Posterior, MLE and MAP

Guiding Question: Naive Bayes fitted itself by counting, and a count of zero vetoed everything. What was counting, really — and what does a prior change?

An Unknown Number, Ten Observations

A new feature went live. Of the first **10 users**, **3 came back**.

What is p, the probability that a user comes back — and how sure are we?

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WHAT THE LAST UNIT CAN SAY

$\hat{p} = 0.3$, standard error ≈ 0.14, a 95% interval of roughly $[0.02, 0.58]$ — about the *procedure*, not about p.

WHAT WE WOULD LIKE TO SAY

"p is probably between 0.1 and 0.6, most likely around 0.3" — a statement **about** p.

Conditioning on the Data — for a Parameter

Treat the unknown p as a random variable θ and apply **Bayes' theorem**, with θ in place of the hypothesis and the data D as the evidence:

$$p(\theta \mid D) = \frac{p(D \mid \theta) \cdot p(\theta)}{p(D)}$$

posterior — belief about θ after the data

likelihood — how probable the data is for each θ

evidence — normalises, does not depend on θ

prior — belief about θ before

For i.i.d. yes/no data with h yeses in n trials: $p(D \mid \theta) = \theta^h(1-\theta)^{n-h}$ — read as a function of θ, with the data fixed.

A Prior That Behaves Like Counts

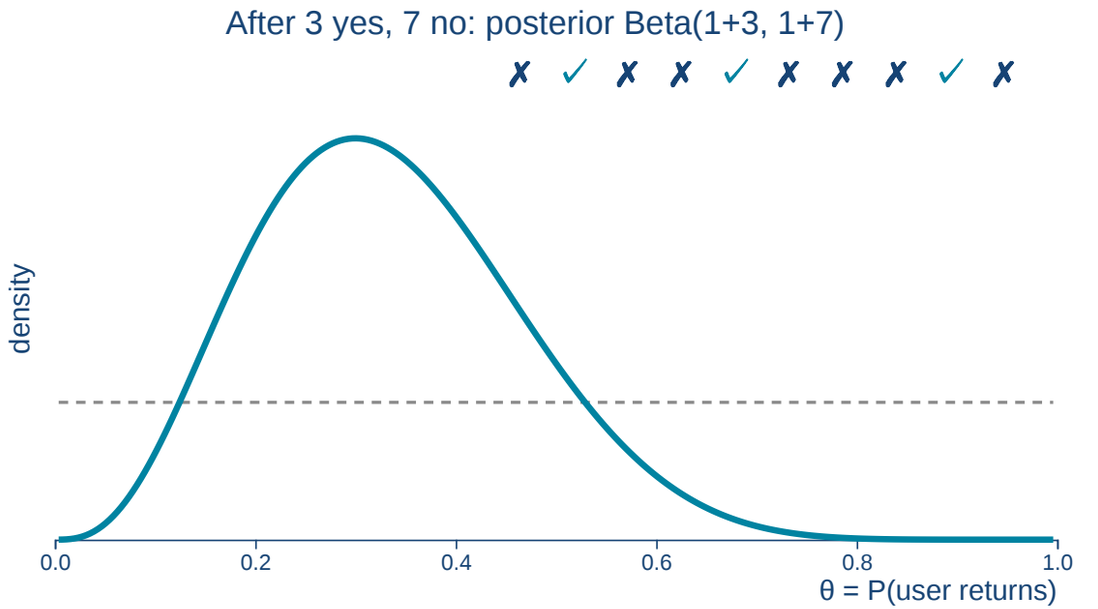
Beta prior. $p(\theta) \propto \theta^{a-1}(1-\theta)^{b-1}$ on $[0,1]$, written $\mathrm{Beta}(a,b)$. $\mathrm{Beta}(1,1)$ is flat: every p equally plausible.

Multiply by the likelihood $\theta^h(1-\theta)^{n-h}$:

$$p(\theta \mid D) \propto \theta^{a+h-1}(1-\theta)^{b+n-h-1} \quad \Longrightarrow \quad \theta \mid D \sim \mathrm{Beta}(a+h,\ \ b+n-h)$$

The update is **adding counts**: a and b act like yeses and noes seen before the data. A prior whose posterior stays in the same family is called **conjugate**.

Watching the Posterior Form



Predict before each click: which way does the curve move after a ✗ — and does it get narrower every time?

What the Posterior Gives You

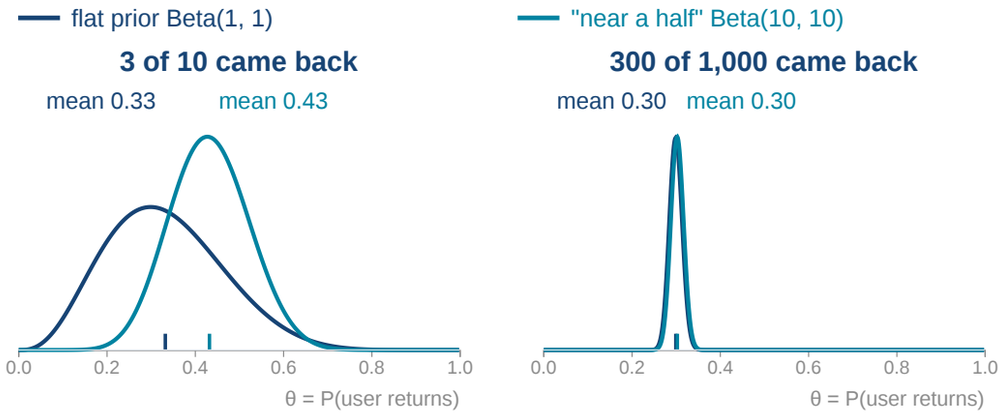
After 3 of 10, with a flat prior: $\theta \mid D \sim \mathrm{Beta}(4, 8)$.

question	answer	from
most plausible p	$0.30 = h/n$	the peak (mode)
best guess on average	0.33	the posterior mean $\frac{a+h}{a+b+n}$
where is p, 95%?	$[0.11,\ 0.61]$	a **credible interval**
after 1,000 users, 300 back	$[0.27,\ 0.33]$	same model, more data

> The peak under a flat prior is exactly h/n. **Lecture 03** keeps only this peak and calls it maximum likelihood — the first shortcut of the course.

How Much Does the Prior Matter?

Two analysts: one with a flat prior $\mathrm{Beta}(1,1)$, one convinced that p is near a half, $\mathrm{Beta}(10,10)$.



> With little data the prior matters; with enough data the analysts **agree**. A prior is a stated assumption — it can be argued with, and it washes out.

Mini-Exercise

Prior $\mathrm{Beta}(2, 2)$. Of 10 users, **8** come back.

Is the posterior mean above or below 0.8 — and why?

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Below. The posterior is $\mathrm{Beta}(2 + 8,\ 2 + 2) = \mathrm{Beta}(10, 4)$, mean $10/14 \approx 0.71$. The prior counts two yeses and two noes, so it pulls towards 0.5 — and the pull fades as data grows.

Two Ways to Answer "Where Is p?"

	random	the 95% interval	needs
Frequentist	the data; p is fixed	a **procedure** that catches p in 95% of samples — about this one interval, no probability statement	nothing but the sampling model
Bayesian	p; the data is fixed	"p lies in this interval with probability 0.95" — a true sentence about p	a **prior**: what we believed about p before the data

> The axioms are the same; the question is not. The rest of the course mostly takes the second road — and Lecture 03 starts with its cheapest version: keep only the peak.

Takeaway

An unknown parameter is just another unknown: give it a **prior**, condition on the data, and the **posterior** says what the data justifies — as a peak, a mean and an interval. For yes/no data with a Beta prior the update is **adding counts**, and the prior's influence fades as data grows.

What Was Counting, Really?

Naive Bayes set $P(\text{new device} \mid \text{fraud}) = 60/100$ and $P(\text{crypto} \mid \text{fraud}) = 0/100$. Which principle produces these numbers — and which one would have avoided the 0?

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The share is the parameter value under which the observed data are **most probable**: the **maximum likelihood estimate**. Smoothing used a prior and took the posterior **mean**.

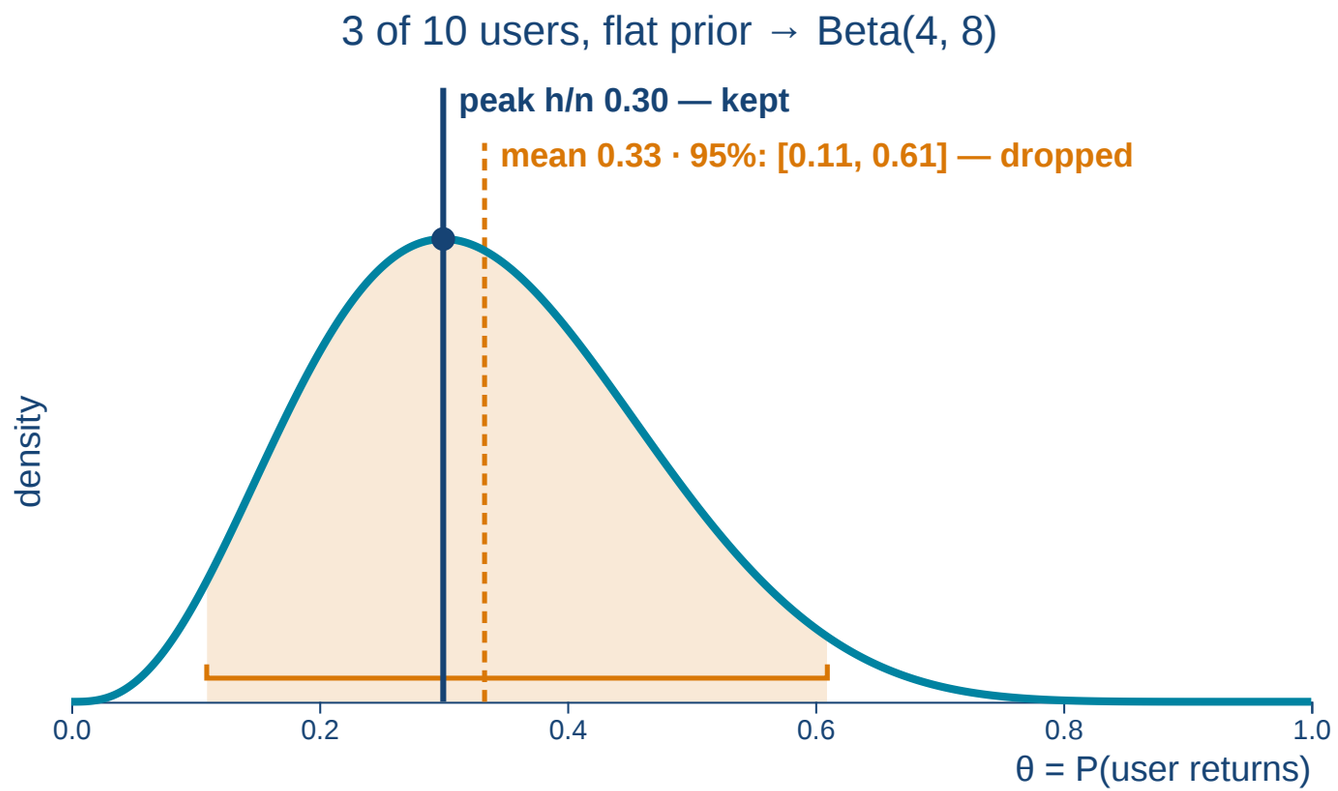
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The share is the parameter value under which the observed data are **most probable**: the **maximum likelihood estimate**. Smoothing used a prior and took the posterior **mean**.

> Both are readings of the posterior over the parameter just built — one its peak, one its mean.

One Posterior, Three Readings



- **peak** $0.30 = h/n$ — the share
- **mean** 0.33 — the smoothed value
- **95% interval** $[0.11, 0.61]$

> Naive Bayes kept only the **peak**. What does that drop — and why does almost everyone do it?

The Likelihood, Read Along the Other Axis

The **likelihood** is the probability of the observed data, read as a function of the **parameter** with the data held fixed: $L(\theta) = p(D \mid \theta)$. The **maximum likelihood estimate** is its peak, $\hat{\theta}_{\mathrm{ML}} = \arg\max_\theta p(D \mid \theta)$.

Lecture 03, 3 of 10 users

$$p(\theta \mid D) \propto \theta^3 (1-\theta)^7 \cdot 1$$

With a **flat prior** the posterior *is* the normalised likelihood — so its peak $h/n = 0.3$ is the maximum likelihood estimate.

Not a distribution over θ

$$\int_0^1 \theta^3 (1-\theta)^7 \mathrm{d}\theta = \frac{1}{1320}$$

The likelihood is a probability over **data**. Read along θ it need not integrate to 1 — only the posterior does.

Deriving h/n Without a Posterior

The i.i.d. assumption of Lecture 02 lets the likelihood factorise over rows; the logarithm turns the product into a sum without moving the peak:

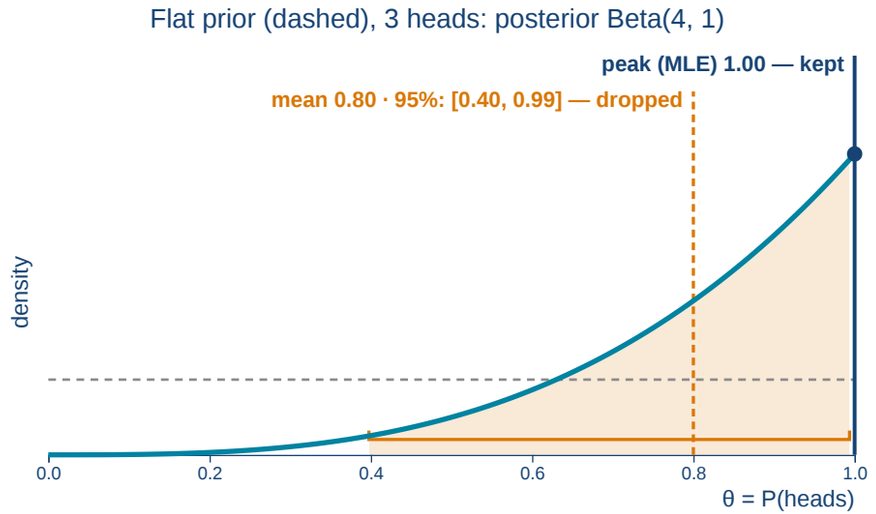
$$\ell(\theta) = \log \prod_{i=1}^{n} \theta^{y_i} (1-\theta)^{1-y_i} = h \log \theta + (n-h) \log(1-\theta)$$

$$\frac{d\ell}{d\theta} = \frac{h}{\theta} - \frac{n-h}{1-\theta} = 0 \quad \Longrightarrow \quad \hat{\theta}_{\mathrm{ML}} = \frac{h}{n}$$

> Last week's peak, found by **setting a derivative to zero —** no prior, no normaliser, no integral. The recipe works unchanged for a million parameters, where nobody can draw the posterior.

The Peak Can Lie

Three flips, **three heads**. Maximum likelihood: $\hat{\theta}_{\mathrm{ML}} = 3/3 = 1$.



Predict before the second click: where do the posterior mean and the 95% interval lie?

A Prior's Peak: MAP

Keep the prior, and still take only the peak — the **maximum a posteriori** estimate. With a $\mathrm{Beta}(a, b)$ prior, the same derivative gives

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} p(D \mid \theta) p(\theta) = \frac{h + a - 1}{n + a + b - 2}$$

prior	data	MLE h/n	MAP	posterior mean
$\mathrm{Beta}(1, 1)$ flat	3 of 3	1	1	0.80
$\mathrm{Beta}(2, 2)$	3 of 3	1	**0.80**	0.71
$\mathrm{Beta}(2, 2)$	8 of 10 (Lecture 03)	0.80	**0.75**	0.71

The prior shifts the peak as if $a - 1$ yeses and $b - 1$ noes had been seen before the data — a **flat** prior shifts nothing. MAP pulls 3 of 3 back to 0.8 — and still reports one number.

Why Everyone Keeps the Peak

	the whole posterior	only the peak
needs	the normaliser $p(D)$ — an integral over θ	a derivative
in 1 dimension	a Beta table	h/n
in a million	an integral nobody can compute exactly	gradient descent
costs	computation	the **spread**

With **1,000 users** the posterior is $\mathrm{Beta}(301, 701)$: standard deviation 0.015. The peak is nearly everything there is to know.

Mini-Exercise

Two analysts both report $\hat{\theta} = 0.3$. One saw **3 of 10** users return, the other **300 of 1,000**.

What does the reported number hide — and which estimate would you bet on?

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The **spread**. The posteriors of Lecture 03: $[0.11, 0.61]$ after 10 users, $[0.27, 0.33]$ after 1,000 — the same peak, an interval nine times wider. Bet on the second; a point estimate cannot tell you to.

Three Choices, Made Separately

choice	options	decides
what to model	generative: $p(y)$, $p(x \mid y)$ · discriminative: $p(y \mid x)$	which distribution is learned
how to fit θ	MLE · MAP · keep the whole posterior	which parameters we keep
how to decide	most probable class · a threshold from the costs	what we do with $P(y \mid x)$

Two different "MAP"s. MAP *estimation* picks the most probable **parameter** given the training data. The MAP *decision* picks the most probable **class** for one input, $\arg\max_y P(y \mid x)$.

Takeaway

Shortcut 1 — keep the peak. Maximum likelihood (flat prior) and MAP (with a prior) keep the posterior's most probable point; naive Bayes' counting was the first. It is cheap — a derivative instead of an integral — and harmless with plenty of data. It drops the **spread**, and it is a choice of its own: generative or discriminative, every model is fitted this way.